

* Statistics

* Mean with only observations

$$\rightarrow \frac{\sum x}{N}$$

* Mean with ungrouped data or with frequency

$$\rightarrow \frac{\sum fx}{N} \rightarrow \frac{\sum fx}{\sum f}$$

* Shortcut Method with observations

$$\rightarrow \frac{\sum fdx}{\sum f} + A = \bar{x}, \quad d = x - A, \quad A = \text{Assumed Mean}$$

* Mean with class Interval

$$\rightarrow \bar{x} = A + \frac{\sum fdx}{\sum f} \times C, \quad C = \text{class Interval}$$

$$d = \frac{x - A}{C}$$

* Combined Mean

$$\bar{x}_C = \frac{N_1 \bar{x}_1 + N_2 \bar{x}_2 + \dots + N_n \bar{x}_n}{N_1 + N_2 + \dots + N_n}$$

* Weighted Mean

$$\bar{x}_W = \frac{\sum W_i \cdot x_i}{\sum W} = \frac{W_1 \cdot x_1 + W_2 \cdot x_2 + \dots + W_n \cdot x_n}{W_1 + W_2 + W_3 + \dots}$$

* Geometric Mean With Observations

$$\bar{x}_G = GM = \text{Antilog } \frac{1}{n} [\sum \log x]$$

If log of any value the length of value

minus 1. ex 0 = 0. } log value
333 = 2. }
12 = 1. }

If Antilog value the point (.) putting in the answer with preceding the plus + 1 in the answer.

ex 0.12. = 1.2 } 3 + 1 = 4
3.1235 = 3123.5 } Antilog value

* Geometric Mean With Frequency

$$\bar{x}_G = \text{Antilog } \left[\frac{\sum \log x f}{n} \right]$$

→ log x → log x x f → $\sum \log x f$

* Geometric Mean With Class Interval

$$\bar{x}_G = \text{Antilog } \left[\frac{\sum \log x f}{n} \right]$$

∴ class interval
10-20

Find mid-value x using class interval.

ex $xc = \frac{10+20}{2} = 15$

* Harmonic Mean with Observations

$$\bar{x}_H = \frac{N}{\sum (1/x)} \quad \rightarrow \text{Number of Observation}$$

* Harmonic Mean with Frequency

$$\bar{x}_H = \frac{\sum f}{\sum (f/x)} \quad \rightarrow \frac{N}{\sum (f/x)}$$

Summation of all frequency

* Harmonic Mean with class Interval

$$\bar{x}_H = \frac{\sum f}{\sum (f/x)} \quad \rightarrow \text{Find mid-value or using class Interval}$$

* Median with Observations

$$M = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ value} \quad \rightarrow \text{Observation}$$

Note: Arrange in Descending Order first.

If Answer in decimal value so you can assume the observation like ex

ex $M = 6.8^{\text{th}} \text{ value} \rightarrow \frac{6^{\text{th}} \text{ obs} + 7^{\text{th}} \text{ obs}}{2} \Rightarrow \boxed{M}$

ex $M = 10.5^{\text{th}} \text{ value} \rightarrow \frac{10^{\text{th}} \text{ obs} + 11^{\text{th}} \text{ obs}}{2} \Rightarrow \boxed{M}$

* Median with frequency

→ $M = \left(\frac{N+1}{2}\right)^{th}$ value in CF observation

→ CF $\Rightarrow$ summation of all frequency after and after.

ex $M = 61.5^{th}$ obser in CF

ex class frequency

x	f	cf
1000	24	40
1500	26	66

→ CF observations

Median

∴ Note: Arrange in Ascending Order

* Median with class interval

$$M = L + \frac{\left(\frac{n}{2}\right) - cf}{f} \times c$$

$$M = \left(\frac{n}{2}\right)^{th} \text{ obser in CF}$$

$$\left(\frac{n}{2}\right) = 25^{th} \text{ obs in CF}$$

ex

Convert class

class	f	cf
40-79	14	16
80-114	23	39

Consider CF

→ obser CF

→ class Interval 80-114 → convert → 79.5-119.5

ex $L = 79.5$, $CF = 16$ & $F = 23$, $(n/2) = 25$

$C = 40 \rightarrow$ difference of class Interval

$$\boxed{79.5 - 119.5}$$

* Mode with observations

$\rightarrow Z \rightarrow$ Repeated value More

ex 2, 3, 3, 3, 2, 5, 6

$$Z = \boxed{3}$$

* Mode with frequency

$\rightarrow Z \rightarrow$ Find highest frequency & their frequency observation is your Mode.

ex ob 12 $\boxed{13} \rightarrow Z$
f 38 52

$\rightarrow$ highest frequency

$$Z = \boxed{13}$$

* Mode with class Interval

$$Z = L + \left(\frac{f_0 - f_1}{2f_0 - f_1 - f_2} \right) \times C \quad \left| \begin{array}{l} \text{Note :- Find} \\ \text{highest frequency} \end{array} \right.$$

ex class
do not convert $\int$ 15-20
20-25
25-30

f
 $\boxed{32} \rightarrow f_1$
 $\boxed{45} \rightarrow f_0$
 $\boxed{27} \rightarrow f_2$

$$\boxed{C = 5}$$

* The data is given all form like

$$\left. \begin{array}{l} 0 \\ 10-20 \\ 30-40 \end{array} \right\} \text{ so use this formula}$$

$$Z = 3M - 2\bar{x}$$

* Quantile with Observation

$$Q_j = j \left(\frac{n+1}{4} \right)^{\text{th}} \text{ obser.} \Rightarrow Q_3 = 3 \left(\frac{n+1}{4} \right)^{\text{th}} \text{ obs}$$

If observation in decimal value
for example

65.5th

Note:- $\Rightarrow 65^{\text{th}} \text{ obs} + 0.5 (66^{\text{th}} \text{ obs} - 65^{\text{th}} \text{ obs})$

Note: $\Rightarrow$ Arrange in Ascending Order.

* Quantile with Frequency

+ we find CF

$$Q_j = j \left(\frac{n+1}{4} \right)^{\text{th}} \text{ obs in CF}$$

The observation of CF that observation is
your Q_j .

$$\begin{array}{ccc} \text{ex} & \begin{array}{c} \text{cf} \\ \boxed{3} \\ \text{Q}_j \end{array} & \begin{array}{c} f \\ 12 \end{array} \begin{array}{c} \text{cf} \\ \boxed{32} \end{array} \rightarrow \text{cf} \end{array}$$

* Quantile with class Interval

$Q_j = j\left(\frac{n}{4}\right)$ th obs in cf

$$Q_j = L + \frac{j\left(\frac{n}{4}\right) - cf}{f} \times c$$

* Percentile with observation

$P_j = j\left(\frac{n+1}{100}\right)$ th obs $\Rightarrow P_4 = 4\left(\frac{n+1}{100}\right)$ th obs
 $\rightarrow$ And same as Quantile

* Percentile with Frequency

$P_j = j\left(\frac{n+1}{100}\right)$ th obs in cf

$\therefore$ And same as Quantile

* Percentile with class Interval

$P_j = j\left(\frac{n}{100}\right)$ th obs in cf

$$P_j = L + \frac{j\left(\frac{n}{100}\right) - cf}{f} \times c$$

$\therefore$ And same as Quantile

* Decile with observations

$$\rightarrow D_j = j \left(\frac{n+1}{10} \right)^{\text{th}} \text{ obs} \Rightarrow D_5 = 5 \left(\frac{n+1}{10} \right)^{\text{th}} \text{ obs}$$

$\therefore$ And same as Quartile.

* Decile with Frequency

$$\rightarrow D_j = j \left(\frac{n+1}{10} \right)^{\text{th}} \text{ obs in CF}$$

$\therefore$ And same as Quartile.

* Decile with class Interval

$$\rightarrow D_j = j \left(\frac{n}{10} \right)^{\text{th}} \text{ obs in CF}$$

$$\rightarrow D_j = L + \frac{j \left(\frac{n}{10} \right) - CF}{f} \times c$$

$\therefore$ And same as Quartile.

* Range with observations

$$R = \text{highest obs} - \text{lowest obs}$$

$$R = H - L$$

$$\text{Co-efficient of } R = \frac{H - L}{H + L}$$

* Range with Frequency

$R =$ highest obs - lowest obs in $\boxed{OG}$

$$R = h - L$$

$$\text{Co-ef of } R = \frac{h - L}{h + L}$$

* Range with Class Interval

$R =$ highest class interval - lowest class interval

$$\text{ex } \boxed{40-46} \quad \dots \quad 54 \text{ } \boxed{54-56}$$

$\downarrow \qquad \qquad \qquad \downarrow$
 $L \qquad \qquad \qquad H$

$$R = H - L$$

$$\text{Co-ef of } R = \frac{H - L}{H + L}$$

* Quantile Deviation

$$Q_d = \frac{Q_3 - Q_1}{2}$$

$$\text{Co-ef of } Q_d = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Note: Same as Quartile

$\therefore Q_3$ & Q_1 find according the observations are given or frequency are given or also class is given.

* Mean Deviation

$$MD = \frac{\sum |x - \bar{x}|}{n} \rightarrow \text{ignore sign}$$

$$\text{co-ef of MD} = \frac{MD}{\bar{x}}$$

$\therefore$ Find $\bar{x}$ According data are given i.e observation are given, or class are given or also frequency are given

* Standard Deviation

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$S = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} \quad \text{or}$$

$$S = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2} \quad \text{or} \quad \boxed{\times C}$$

$$S = \sqrt{\frac{\sum f x^2}{\sum f} - \left(\frac{\sum f x}{\sum f}\right)^2} \quad \text{or}$$

$$S = \sqrt{\frac{\sum f d^2}{\sum f} - \left(\frac{\sum f d}{\sum f}\right)^2} \quad \text{or}$$

Note: According data you choose which formula are best.

$\rightarrow$ use only observation

If class are given
So you multiply
class interval

$\rightarrow$ use class are given

Note: $\sum f x^2$

$\rightarrow$

$f x \rightarrow f x$

$f x \times x \rightarrow f x^2$

* Variance

$$V = \sigma^2 \rightarrow \sigma \rightarrow \text{Standard deviation}$$

* Co-efficient of SD (Standard deviation)

$$\rightarrow \text{co. ef of SD} = \frac{\sigma}{\bar{x}}$$

$$* \text{ Co-efficient of Variance} \Rightarrow \frac{\sigma}{\bar{x}} \times 100$$

* Combined Standard Deviations

$$S_d = \sqrt{\frac{n_1(s_1^2 + d_1^2) + n_2(s_2^2 + d_2^2)}{n_1 + n_2}}$$

$$d_1 = \bar{x}_1 - \bar{x}_c \quad , \quad d_2 = \bar{x}_2 - \bar{x}_c$$

↓
Combined
Mean

$$\bar{x}_c = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$